

# Topic 3: Asymptotics

02-680: Essentials of Mathematics and Statistics

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## 1 Big O

To understand function behavior of say function  $f(x)$  we often want to look at its growth compared to another function  $g(x)$ , for instance we might ask:

Does  $f(x)$  grow no faster than  $g(x)$ ?

Define the **set** of functions  $f(x)$  for which this is true:

For functions  $h : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$  and  $g : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$

$$O(g(x)) := \{h(x) \mid \exists x_0 \geq 0, c > 0 : \forall x \geq x_0 : h(x) \leq c \cdot g(x)\}$$

(notice the explicit use of  $\geq$  vs  $>$ )

## Useful References

Liben-Nowell, “Connecting Discrete Mathematics and Computer Science, 2e”. §6.2